Honors Algebra

MATH-SHU 348

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8/28 Lecture

1 Group Theory

1.1 Permutations of a Set

Definition 1.1. Let X be a set. $f : X \to X$ is a **bijection** if $\forall y \in X$, there exists a unique $x \in X$, such that $f(x) = y$.

Remark 1.2. We observe that $f : X \to X$ is bijective if and only if there exists $g : X \to X$, such that $f \circ g = g \circ f = \mathrm{Id}_X$. Such a mapping g is unique when it exists, denoted as f^{-1}, the inverse mapping of f. We commonly write $\mathfrak{S}(X)$ or $\mathfrak{S}_X$ as the set of bijections from X to X, namely the set of **permutations** of X. Specifically, $\mathfrak{S}_n := \mathfrak{S}(\{1, \cdots, n\})$.

Definition 1.3. Let G be a set, and we say that $\cdot$ is a **binary operation** on G if $\forall x, y \in G$, we have that $x \cdot y \in G$.

1.2 Groups

Definition 1.4. A couple $(G, \cdot)$, where G is a set and $\cdot$ is a binary operation on G, is called a **group** if the operation $\cdot$ satisfies:

(1) Associativity: $\forall g, h, f \in G$, one has $(g \cdot h) \cdot f = g \cdot (h \cdot f)$.

(2) Identity/neutral element: $\exists e \in G$, such that $e \cdot g = g \cdot e = g$, $\forall g \in G$.

(3) Inverse: $\forall g \in G$, $\exists h \in G$, such that $g \cdot h = h \cdot g = e$, *i.e.*, every element in G has an inverse in G.

Remark 1.5. Suppose $(G, \cdot)$ is a group with neutral element e and e', we can see that $e = e \cdot e' = e'$, so that the neutral element e is unique. We denote it by e_G or $\mathbb{1}_G$. Now take $g \in G$ and suppose h and h' are both inverse of g. We can show that

$$h = h \cdot e_G = h \cdot (g \cdot h') = h \cdot g \cdot h' = (h \cdot g) \cdot h' = e_G \cdot h' = h', \tag{1}$$

and thus the inverse is also unique, denoted as $h = g^{-1}$. When $(G, \cdot)$ is a group, the group operation $\cdot$ can also be called as group law or the multiplication on G.

Theorem 1.6. (1) Let $(G, \cdot)$ be a group and $g \in G$, then $(g^{-1})^{-1} = g$.

(2) Take $g, h \in G$ where $(G, \cdot)$ is a group, then $(g \cdot h)^{-1} = h^{-1} \cdot g^{-1}$.

Proof. (1) We can easily see that

$$g \cdot g^{-1} = g^{-1} \cdot g = e_G \implies (g^{-1})^{-1} = g. \tag{2}$$

(2) We can easily see that

$$(h^{-1} \cdot g^{-1}) \cdot (g \cdot h) = h^{-1} \cdot (g^{-1} \cdot g) \cdot h = h^{-1} \cdot e_G \cdot h = h^{-1} \cdot h = e_G, \tag{3}$$

so we can conclude that $(h^{-1} \cdot g^{-1}) = (g \cdot h)^{-1}$.

□

Example 1.7. (1) $(\mathbb{Z}, +)$ is a group with neutral element 0, and inverse "n^{-1}" $= -n$.

(2) $(\mathbb{N}, +)$ is not a group, because $1 \in \mathbb{N}$ does not have an inverse in $\mathbb{N}$ for $+$.

(3) $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, $(\mathbb{Q}, +)$ are groups.

(4) $(\mathbb{R}^*, \times)$, $(\mathbb{C}^*, \times)$, $(\mathbb{Q}^*, \times)$ are groups, where $K^* = K \setminus \{0\}$. We explicitly show that $(\mathbb{Q}^*, \times)$ is a group. To see this, take arbitrary elements $\frac{a}{b}$ and $\frac{c}{d}$ in $\mathbb{Q}^*$ where $a, b, c, d \in \mathbb{Z}^*$. Then $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}^*$. Also, $\times$ is clearly associative, 1 is the neutral element, and there exists the unique inverse element $\frac{b}{a} \in \mathbb{Q}^*$ for any $\frac{a}{b} \in \mathbb{Q}^*$.

(5) $(\mathbb{Z}^*, \times)$ is not a group, since 2 does not have an inverse in $\mathbb{Z}^*$.

(6) Let X be a set, then $(\mathfrak{S}(X), \circ)$ is a group. Indeed, $\circ$ satisfies associativity, the neutral element is Id_X, and the inverse of a mapping f is f^{-1}. It suffices to validate that $\circ$ is a binary operation. Indeed, we have that

$$(f \circ g) \circ (g^{-1} \circ f^{-1}) = f \circ (g \circ g^{-1}) \circ f^{-1} = f \circ f^{-1} = \mathrm{Id}_X, \tag{4}$$

and similarly $(g^{-1} \circ f^{-1}) \circ (f \circ g) = \mathrm{Id}_X$. This means that $f \circ g$ is a bijection when f and g are both bijections, and thus $\circ$ is a binary operator.

(7) $(\mathfrak{M}_{n,m}(K), +)$ is a group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

(8) Let $GL_n(K) := \{\text{invertible matrices in } \mathfrak{M}_n(K)\}$, then $(GL_n(K), \times)$ is a group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

Definition 1.8. A group $(G, \cdot)$ is called **commutative** or **Abelian** if $\forall g, h \in G$, $g \cdot h = h \cdot g$.

Example 1.9. (1) $(\mathbb{Z}, +)$, $(\mathbb{R}^*, \times)$, $(\mathbb{C}^*, \times)$, $(\mathbb{Q}^*, \times)$ are Abelian groups.

(2) $(\mathfrak{M}_{n,m}(K), +)$ is an Abelian group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

(3) $(GL_n(K), \times)$ is not commutative for $n \geq 2$, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$. Clearly high-dimensional matrix multiplication is not commutative.

(4) $(\mathfrak{S}(X), \circ)$ is not commutative for $|X| \geq 3$.

8/30 Lecture

1.3 Subgroups

Definition 1.10. Let $(G, \cdot)$ be a group and let $H \subseteq G$. Then H is a **subgroup** of G if:

(1) $e_G \in H$.

(2) $\forall x, y \in H$, $x \cdot y \in H$.

(3) $\forall x \in H$, $x^{-1} \in H$.

Often, one writes G instead of $(G, \cdot_G)$ and H instead of $(H, \cdot_H)$. We denote the subgroup relation by $H \leq G$.

Remark 1.11. We can easily observe the following. If $(G, \cdot_G)$ is a group and $H \leq G$, then $(H, \cdot_G)$ is a group, $e_H = e_G$, and the inverse of an element in H is equal to its inverse in G. These follow trivially from the definition, the uniqueness of neutral element, and the uniqueness of inverse.

Example 1.12. (1) $(\mathbb{R}, +) \leq (\mathbb{C}, +)$. Indeed, we can check that $0 \in \mathbb{R}$ where 0 is the neutral element of $(\mathbb{C}, +)$. Also, $\forall x, y \in \mathbb{R}$, $x + y \in \mathbb{R}$ and $-x \in \mathbb{R}$.

(2) $\mathbb{C}^*$ is not a subgroup of $\mathbb{C}$, since the neutral element of $\mathbb{C}$, 0, is not in $\mathbb{C}^*$.

(3) For $a \in \mathbb{Z}$, $a\mathbb{Z} := \{ak;\ k \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

(4) For any group G, $\{e_G\}$ and G are clearly subgroups of G. $\{e_G\}$ is called the trivial subgroup of G.

Example 1.13. $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$, and let $GL_n(K) = \{\text{invertible matrices in } \mathfrak{M}_n(K)\}$. Set

$$SL_n(K) = \{M \in GL_n(K);\ \det(H) = 1\}, \tag{5}$$

$$O_n(K) = \left\{M \in GL_n(K);\ M^\top M = I_n\right\}. \tag{6}$$

(1) Show that $SL_n(K) \leq GL_n(K)$. Indeed, the neutral element of $GL_n(K)$ is the n-dimensional identity matrix I_n. Clearly $\det(I_n) = 1$, so that $I_n \in SL_n(K)$. Moreover, taking arbitrary $A, B \in SL_n(K)$, we have that $\det(AB) = \det(A)\det(B) = 1$ and $\det(A^{-1}) = (\det(A))^{-1} = 1$, so that $AB, A^{-1} \in SL_n(K)$. The proof is done.

(2) Show that $O_n(K) \leq SL_n(K)$. Indeed, we have checked that the neutral element of $SL_n(K)$ is I_n, and clearly $I_n^\top I_n = I_n$. Therefore, $I_n \in O_n(K)$. Moreover, taking arbitrary $A, B \in O_n(K)$, we have that $(AB)^\top AB = B^\top(A^\top A)B = B^\top B = I_n$. Since $A^\top A = I_n$, we have that $A^{-1} = A^\top$. Taking transpose on both sides, we can see that $(A^{-1})^\top = (A^\top)^\top = A$. By multiplying A^{-1} from the right on both sides, we can conclude that $(A^{-1})^\top A^{-1} = AA^{-1} = I_n$. The proof is done.

Lemma 1.14. Let $(G, \cdot)$ be a group. If $K \leq G$ and $H \leq K$, then $H \leq G$.

Proof. We have observed that the neutral element of a group and its subgroup is the same, so that $e_H = e_K = e_G \in H$. Now take arbitrary $x, y \in H$, we have that $x \cdot y \in H$ and $x^{-1} \in H$ because $H \leq K$ (for the inverse it is important to also note that the inverse of x in H and G are equal). Therefore it follows that $H \leq G$, and the proof is complete. □

Lemma 1.15. If $K \leq G$, $H \leq G$, and $H \subseteq K$, then $H \leq K$.

Proof. Similar to above, we can see that $e_H = e_G = e_K \in H$. Take arbitrary $x, y \in H$, then $x \cdot y \in H$ and $x^{-1} \in H$ because $H \leq G$. Therefore it follows that $H \leq K$, and the proof is complete. □

Proposition 1.16. Let $(G, \cdot)$ be a group and $H_i \leq G$, $\forall i \in I$, where I is a (possibly infinite) indexing set. Then $\bigcap_{i \in I} H_i \leq G$.

Proof. We have that $e_G \in H_i$, $\forall i \in I$, so that $e_G \in \bigcap_{i \in I} H_i$. Moreover, taking arbitrary $x, y \in \bigcap_{i \in I} H_i$, $x, y \in H_i$, $\forall i \in I$. Then $x \cdot y, x^{-1} \in H_i$, $\forall i \in I$. This implies that $x \cdot y, x^{-1} \in \bigcap_{i \in I} H_i$, so the proof is complete. □

1.4 Groups Generated by a Subset

Definition 1.17. Let $(G, \cdot)$ be a group and $S \subseteq G$. One sets

$$\langle S \rangle = \bigcap_{H \leq G, S \subseteq H} H = \{\text{intersection of all subgroups containing } S\}, \tag{7}$$

where $\langle S \rangle$ is called the **subgroup generated by** S.

Remark 1.18. We observe that $\langle S \rangle \leq H$ by the previous proposition, and $\langle S \rangle$ is in fact the smallest subgroup of G containing S.

Lemma 1.19. Let $(G, \cdot)$ be a group and $H \leq G$. Then $S \subseteq H$ if and only if $\langle S \rangle \subseteq H$.

Proof. $\Longrightarrow$ H is a subgroup of G containing S, and $\langle S \rangle$ is the smallest subgroup of G containing S. Therefore, $\langle S \rangle \subseteq H$ trivially. In fact, $\langle S \rangle \leq H$ by Proposition 1.16.

$\Longleftarrow$ Clearly, $S \subseteq \langle S \rangle$ so this direction is trivial. □

Proposition 1.20. Let $(G, \cdot)$ be a group and $S \subseteq G$, then

$$\langle S \rangle = \{e_G\} \cup \{s_1^{\epsilon_1} \cdots s_n^{\epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1,\ s_i \in S\}. \tag{8}$$

Proof. $\subseteq$ Denote the right-hand side by A. We need to check that A is a subgroup of G containing S, so it will definitely contain the smallest subgroup of G containing S, *i.e.*, $\langle S \rangle$. Clearly for each $s \in S$, we can take $n = 1$ and $s = s_1^{\epsilon_1}$ with $s_1 = s$ and $\epsilon_1 = 1$, so $S \subseteq A$. Now it suffices to check that $A \leq G$. By definition $e_G \in A$. Take arbitrary $a, b \in A$, we can see that $a \cdot b \in A$. This can be discussed in two cases. If one of the elements is e_G, then the product is simply the other element. Otherwise, the two sequences of products can be concatenated into a longer sequence of product, still falling into A. Finally $a^{-1} \in A$, because clearly $e_G^{-1} = e_G \in A$, and for any other element, we can simply revert the signs of the ϵ_i's which falls back to A. Thus we have shown that A is a subgroup of G containing S, and this direction is done.

$\supseteq$ Since $\langle S \rangle \leq G$, we have that $e_G \in \langle S \rangle$. Moreover, each $s_i \in S$, so all their inverses and all possible products of their inverses and themselves are in $\langle S \rangle$. This direction is thus done. □

9/1 Recitation

Example 1.21. For n a positive integer, count the number of elements in $\mathfrak{S}_n$.

Solution. $n!$.

Example 1.22. Let $(G, \cdot)$ be a group and let $x \in G$. For $k \geq 1$, put $x^k = \underbrace{x \cdots x}_{k\times}$. Put $x^0 := 1$ and for $k < 0$ put $x^k := (x^{-1})^{-k}$. Finally let $\langle x \rangle := \langle \{x\} \rangle$. These are standard notations to be always used later.

(1) Check that $x^a \cdot x^b = x^{a+b}$, $\forall a, b \in \mathbb{Z}$.

(2) Check that $\langle x \rangle = \left\{x^k;\ k \in \mathbb{Z}\right\}$.

Proof. The first equality is trivial but discussing each case separately. As for the second equality, we have that

$$\begin{aligned} \langle x \rangle = \langle \{x\} \rangle &= \{1\} \cup \{x_1^{\epsilon_1} \cdots x_n^{\epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1,\ x_i \in \{x\}\} && (9) \\ &= \{1\} \cup \left\{x^{\epsilon_1 + \cdots + \epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1\right\} = \left\{x^k;\ k \in \mathbb{Z}\right\}, && (10) \end{aligned}$$

so the proof is complete. □

Example 1.23. Let G be a group such that $x^2 = e$ for all $x \in G$.

(1) Check that G is commutative.

(2) Suppose that such a group G is a subgroup of $GL_n(\mathbb{R})$. Show that every element in G is diagonalizable with eigenvalues ± 1.

(3) Set $H_n := \{\mathrm{diag}(\epsilon_1, \cdots, \epsilon_n);\ \epsilon_i = \pm 1\} \subseteq GL_n(\mathbb{R})$. With the same assumption as above, show that $H_n \leq G$.

(4) With the same assumptions as above, prove that there exists $P \in GL_n(\mathbb{R})$, such that for every $M \in G$, $PMP^{-1} \in H_n$ (this can be written as $PGP^{-1} \subseteq H_n$ in short).

Proof. (1) $\forall a, b \in G$, we have that $ab \in G$. Therefore, $(ab)^2 = abab = e$. Multiplying by a on the left and b on the right on both sides of the equation, we can see that $ba = a^2bab^2 = ab$, so G is commutative.

(2) Since for each $A \in G$, $A^2 = I$. This means that the minimal polynomial is $\lambda^2 - 1$ which has two distinct roots, so A is diagonalizable. Now suppose $A\mathbf{v} = \lambda\mathbf{v}$. Multiplying by A on the left on both sides of the equation, we have that $\mathbf{v} = A^2\mathbf{v} = \lambda A\mathbf{v} = \lambda^2\mathbf{v}$. This implies that $\lambda = \pm 1$, so the proof is complete.

(3) The neutral element of H_n is $I \in G$. Take arbitrary $A = \mathrm{diag}(a_1, \cdots, a_n) \in H_n$ and $B = \mathrm{diag}(b_1, \cdots, b_n) \in H_n$. Clearly $AB = \mathrm{diag}(a_1b_1, \cdots, a_nb_n) \in H_n$ because $a_ib_i = \pm 1$, and $A^{-1} = \mathrm{diag}(a_1^{-1}, \cdots, a_n^{-1}) \in H_n$ because $a_i^{-1} = \pm 1$. Hence we can conclude that $H_n \leq G$.

(4) To be done... □

9/4 Lecture

1.5 Group Homomorphisms

Definition 1.24. Let $(G, \cdot)$ and $(G', *)$ be two groups. A mapping $f : G \to G'$ is called a **group homomorphism** if $f(x \cdot y) = f(x) * f(y)$, $\forall x, y \in G$.

Remark 1.25. If $f : G \to G'$ is a group isomorphism, then $f(e) = e'$. Indeed, we can see that $f(e) = f(e \cdot e) = f(e) * f(e)$. Multiplying on the left by $(f(e))^{-1}$ on both sides of the equation, we have that $e' = f(e)$. Moreover, $f(x^{-1}) = (f(x))^{-1}$, $\forall x \in G$. Indeed, we can see that $e' = f(e) = f(x \cdot x^{-1}) = f(x) * f(x^{-1})$, so $f(x^{-1}) = (f(x))^{-1}$.

Remark 1.26. We denote the set of homomorphisms from G to G' by $\mathrm{Hom}(G, G')$. We further define the set of **endomorphisms** on G to be $\mathrm{End}(G) := \mathrm{Hom}(G, G)$, the homomorphisms from G to iteself. For $f \in \mathrm{Hom}(G, G')$, if it is furthermore bijective, we say that $f \in \mathrm{Iso}(G, G')$, the set of **isomorphisms** from G to G'. In what follows, we write the group law $\cdot$ for abstract groups. In particular, writing $(G, \cdot)$ and $(G', \cdot)$ does not mean that $\cdot_G = \cdot_{G'}$. It is an abuse of notation and may sometimes be omitted.

Example 1.27. (1) $\det : GL_n(K) \to K^*$ belongs to $\mathrm{Hom}(GL_n(K), K^*)$. Indeed, $\det(AB) = \det(A)\det(B)$.

(2) For a fixed $a \in \mathbb{Z}$, the mapping $m_a : \mathbb{Z} \to \mathbb{Z}$ such that $x \mapsto ax$, belongs to $\mathrm{End}(\mathbb{Z})$. Indeed, $m_a(x+y) = a(x+y) = ax + ay = m_a(x) + m_a(y)$. CHECK THAT $\mathrm{End}(\mathbb{Z}) = \{m_a;\ a \in \mathbb{Z}\}$. CHECK THAT if $f \in \mathrm{End}(\mathbb{Z})$, then $f = m_{f(1)}$.

(3) If G and G' are groups, then the **trivial isomorphism** $\mathbb{1}_{G,G'} : G \to G'$ defined by $\mathbb{1}_{G,G'}(x) = e'$, $\forall x \in G$, is a group homomorphism from G to G'. In particular, $\mathrm{Hom}(G, G') \neq \varnothing$ because it always contains $\mathbb{1}_{G,G'}$.

(4) $\mathrm{Id}_{G,G'} : x \mapsto x$ is a endomorphism of G. In particular, $\mathrm{End}(G)$ contains at least two elements $\mathbb{1}_{G,G}$ and Id_G as long as $G \neq \{e\}$.

Proposition 1.28. If $f_1 \in \mathrm{Hom}(G_1, G_2)$ and $f_2 \in \mathrm{Hom}(G_2, G_3)$, then $f_2 \circ f_1 \in \mathrm{Hom}(G_1, G_3)$.

Proof. Take arbitrary $x, y \in G_1$, then we have that

$$(f_2 \circ f_1)(x \cdot y) = f_2(f_1(x \cdot y)) = f_2(f_1(x) \cdot f_1(y)) = f_2(f_1(x)) \cdot f_2(f_1(y)) = (f_2 \circ f_1)(x) \cdot (f_2 \circ f_1)(y), \tag{11}$$

so the proof is complete. □

Proposition 1.29. For $f \in \mathrm{Hom}(G, G')$, if $H \leq G$ and $H' \leq G'$, then $f(H) \leq G'$ and $f^{-1}(H') \leq G$.

Proof. Since $e \in H$, $e' = f(e) \in f(H)$. Take arbitrary $x', y' \in f(H)$, then there exists $x, y \in H$, such that $f(x) = x'$ and $f(y) = y'$. Clearly $(x')^{-1} = (f(x))^{-1} = f(x^{-1}) \in f(H)$ because $x^{-1} \in H$. Also, $x' \cdot y' = f(x) \cdot f(y) = f(x \cdot y) \in f(H)$ because $x \cdot y \in H$. Up till now we have shown that $f(H) \leq G$. Similarly, since $e' = f(e) \in H'$, $e \in f^{-1}(H')$. Take arbitrary $x, y \in f^{-1}(H')$, then $f(x), f(y) \in H'$. Clearly $x^{-1} \in f^{-1}(H')$ because $f(x^{-1}) = (f(x))^{-1} \in H'$. Also, $x \cdot y \in f^{-1}(H')$ because $f(x \cdot y) = f(x) \cdot f(y) \in H'$. Now we have also shown that $f^{-1}(H') \leq G$, so the proof is complete. □

Definition 1.30. For $f \in \mathrm{Hom}(G, G')$, one sets $\mathrm{Ker}(f) := \{x \in G;\ f(x) = e'\} = f^{-1}(\{e'\})$ as the **kernel** of f, and $\mathrm{Im}(f) := f(G)$ as the **image** of f.

Example 1.31. (1) $\mathrm{Im}(\det) = K^*$ and $\mathrm{Ker}(\det) = SL_n(K)$, where $\det \in \mathrm{Hom}(GL_n(K), K^*)$. Recall that $SL_n(K)$ is given by $SL_n(K) = \{M \in GL_n(K);\ \det(H) = 1\}$.

(2) $\mathrm{Im}(\exp) = \mathbb{R}_{>0}$ and $\mathrm{Ker}(\exp) = \{0\}$, where $\exp \in \mathrm{Hom}(\mathbb{R}, \mathbb{R}_{>0})$ is the real exponential mapping.

(3) $\mathrm{Im}(\exp) = \mathbb{C}^*$ and $\mathrm{Ker}(\exp) = 2i\pi\mathbb{Z}$, where $\exp \in \mathrm{Hom}(\mathbb{C}, \mathbb{C}^*)$ is the complex exponential mapping.

Proposition 1.32. If $f \in \mathrm{Hom}(G, G')$, then $\mathrm{Ker}(f) \leq G$ and $\mathrm{Im}(f) \leq G'$.

Proof. This is a direct consequence of Proposition 1.29. □

Proposition 1.33. If $f \in \mathrm{Hom}(G, G')$, then f is injective if and only if $\mathrm{Ker}(f) = \{e\}$, and f is surjective if and only if $\mathrm{Im}(f) = G$.

Proof. The second point is obvious, so we only prove the first point. If f is injective, then there can only be a unique element mapped to e'. Since $f(e) = e'$, we have that $\mathrm{Ker}(f) = \{e\}$. Conversely assume that $\mathrm{Ker}(f) = \{e\}$. Take arbitrary $x, y \in G$ such that $f(x) = f(y)$, then $f(x \cdot y^{-1}) = f(x) \cdot (f(y))^{-1} = e'$. This implies that $x \cdot y^{-1} = e$ since $\mathrm{Ker}(f) = \{e\}$. But then by uniqueness of inverse, $x = y$, so f is indeed injective. □

Proposition 1.34. If $f \in \mathrm{Iso}(G, G')$, then $f^{-1} \in \mathrm{Iso}(G', G)$.

Proof. Take arbitrary $x', y' \in G'$, then since f is a bijection, we can set $x = f^{-1}(x')$ and $y = f^{-1}(y')$. Then $f^{-1}(x' \cdot y') = f^{-1}(f(x) \cdot f(y)) = f^{-1}(f(x \cdot y)) = x \cdot y = f^{-1}(x') \cdot f^{-1}(y')$, which implies that $f^{-1} \in \mathrm{Hom}(G', G)$. Clearly f^{-1} is a bijection because f is a bijection, so $f^{-1} \in \mathrm{Iso}(G', G)$ and the proof is complete. □

Definition 1.35. We say that two groups G and G' are **isomorphic** if $\mathrm{Iso}(G, G') \neq \varnothing$.

Example 1.36. The real exponential mapping $\exp : \mathbb{R} \to \mathbb{R}_{>0}$ belongs to $\mathrm{Iso}(\mathbb{R}, \mathbb{R}_{>0})$. This amounts to say that studying $(\mathbb{R}_{>0}, \times)$ as a group is equivalent to studying $(\mathbb{R}, +)$ as a group, which is more familiar.

9/6 Lecture

1.6 Automorphisms

Definition 1.37. Let $\mathrm{Aut}(G) := \mathrm{Iso}(G, G)$. We call an element in $\mathrm{Aut}(G)$ an **automorphism** of G.

Proposition 1.38. We have that $(\mathrm{Aut}(G), \circ) \leq (\mathfrak{S}(G), \circ)$.

Proof. Clearly the neutral element $\mathrm{Id}_G \in \mathrm{Aut}(G)$. Take arbitrary $f, g \in \mathrm{Aut}(G)$, then $f \circ g \in \mathrm{Aut}(G)$ by Proposition 1.28 since automorphism are homomorphisms. Also, $f^{-1} \in \mathrm{Aut}(G)$ by Proposition 1.34 since automorphisms are isomorphisms. Therefore $(\mathrm{Aut}(G), \circ) \leq (\mathfrak{S}(G), \circ)$, so the proof is complete. □

Example 1.39. $\mathrm{Aut}(\mathbb{Z}) = \{m_1, m_{-1}\} = \{\pm \mathrm{Id}_{\mathbb{Z}}\}$. Indeed, $\forall m \in \mathbb{Z}$, we have that $f(m) = f(1) + \cdots + f(1) = f(1)m$, which means that any $f \in \mathrm{Aut}(\mathbb{Z})$ must be such that $m \mapsto f(1)m$. Now f must also be bijective. Suppose $f(1)m = 1$, then since $f(1), m \in \mathbb{Z}$, the only possibilities are $f(1) = \pm 1$. Therefore, either $m \mapsto m$ or $m \mapsto -m$, and we can check that they are indeed automorphisms of $\mathbb{Z}$.

Remark 1.40. For x in a group G, we define $\iota_x : G \to G$, such that $g \mapsto x \cdot g \cdot x^{-1}$. Then for $x \in G$, we can see that $\iota_x \in \mathrm{Aut}(G)$. Indeed, take arbitrary $f, g \in G$, we have that

$$\iota_x(f \cdot g) = x \cdot (f \cdot g) \cdot x^{-1} = x \cdot f \cdot (x^{-1} \cdot x) \cdot g \cdot x^{-1} = (x \cdot f \cdot x^{-1}) \cdot (x \cdot g \cdot x^{-1}) = \iota_x(f) \cdot \iota_x(g). \tag{12}$$

To see that ι_x is bijective, we claim that $\iota_{x^{-1}}$ is its inverse. Indeed, we can see that

$$(\iota_x \circ \iota_{x^{-1}})(f) = \iota_x(\iota_{x^{-1}}(f)) = \iota_x(x^{-1} \cdot f \cdot x) = x \cdot (x^{-1} \cdot f \cdot x) \cdot x^{-1} = f, \qquad \forall f \in G. \tag{13}$$

The case is the same for $\iota_{x^{-1}} \circ \iota_x$, both equal to Id_G. This implies that ι_x is bijective, and an automorphism of G.

Remark 1.41. The mapping $\iota : G \to \mathrm{Aut}(G)$ such that $x \mapsto \iota_x$ is a group homomorphism. Indeed, take arbitrary $x, y \in G$, we can check that

$$\begin{aligned} \iota(x \cdot y)(g) &= \iota_{x \cdot y}(g) = (x \cdot y) \cdot g \cdot (x \cdot y)^{-1} = x \cdot (y \cdot g \cdot y^{-1}) \cdot x^{-1} \\ &= x \cdot \iota_y(g) \cdot x^{-1} = \iota_x(\iota_y(g)) = (\iota_x \circ \iota_y)(g) = (\iota(x) \circ \iota(y))(g), \qquad \forall g \in G. \end{aligned} \tag{14}$$

Definition 1.42. We set $\mathrm{Inn}(G) := \iota(G)$. An element in $\mathrm{Inn}(G)$ is called an **inner automorphism** of G. We call ι_x **the conjugation by** x.

Definition 1.43. Two elements $x, y \in G$ are said to be **conjugate** if there exists $g \in G$, such that $y = \iota_g(x) = gxg^{-1}$.

Proposition 1.44. Conjugation on the group G, denoted $\sim$, is an equivalence relation.

Proof. We have that $\iota_x(x) = xxx^{-1} = x$ so $x \sim x$, which means that conjugation is reflexive. If $x \sim y$ and $y \sim z$, we assume that $y = \iota_f(x)$ and $z = \iota_g(y)$. Then $x = \iota_f^{-1}(y) = \iota_{f^{-1}}(y)$ so $y \sim x$, which means that conjugation is symmetric. Moreover, $z = \iota_g(\iota_f(x)) = \iota_{gf}(x)$ so $x \sim z$, which means that conjugation is transitive. These three properties show that conjugation on the group G is an equivalence relation. □

Remark 1.45. We denote by $Cl_G(x)$ the **conjugacy class** of x in G.

Lemma 1.46. Take σ and $c = (a_1, \cdots, a_m)$ in $\mathfrak{S}_n$. Then

$$\sigma \circ c \circ \sigma^{-1} = \iota_\sigma(c) = (\sigma(a_1), \cdots, \sigma(a_m)). \tag{15}$$

Proof. Note that c here is a specific type of permutation where elements are cyclically permuted within the cycle, and all other elements are fixed. There are thus two cases.

- If $x \notin \{\sigma(a_1), \cdots, \sigma(a_m)\}$, then $\sigma^{-1}(x) \notin \{a_1, \cdots, a_m\}$. Therefore, $\sigma^{-1}(x)$ remain fixed when permuted, *i.e.*, $c(\sigma^{-1}(x)) = \sigma^{-1}(x)$. This means that $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(\sigma^{-1}(x)) = x$. Indeed, in this case x should be fixed under the cyclic permutation $(\sigma(a_1), \cdots, \sigma(a_m))$, so we are done.
- If $x \in \{\sigma(a_1), \cdots, \sigma(a_m)\}$, then $\sigma^{-1}(x) \in \{a_1, \cdots, a_m\}$. Therefore, $\sigma^{-1}(x)$ will be cyclically permuted within the cycle $(a_1, \cdots, a_m)$. Suppose $x = \sigma(a_i)$, $1 \leq i \leq m-1$. Then $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(c(a_i)) = \sigma(a_{i+1})$. Indeed, $x = \sigma(a_i)$ will be cyclically permuted as $\sigma(a_{i+1})$ under $(\sigma(a_1), \cdots, \sigma(a_m))$ when $1 \leq i \leq m$. Otherwise, $x = \sigma(a_m)$. Then $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(c(a_m)) = \sigma(a_1)$. Indeed, $x = \sigma(a_m)$ will be cyclically permuted as $\sigma(a_1)$ under $(\sigma(a_1), \cdots, \sigma(a_m))$. We have checked all cases up till now, so the proof is complete. □

9/8 Recitation

Example 1.47. Let $G \leq \mathbb{Z}$ and suppose that $G \neq \{0\}$.

(1) Prove that the positive integer $m := \min(G \cap \mathbb{N} \setminus \{0\})$ is well-defined.

(2) Prove that $G = m\mathbb{Z}$.

Hence the only subgroups of $\mathbb{Z}$ are the groups $m\mathbb{Z}$ for $m \in \mathbb{N}$.

Proof. (1) We recall the following theorems. *$S \subseteq \mathbb{N}$ has a minimum if and only if $S \neq \varnothing$; $S \subseteq \mathbb{Z}$ has a minimum if and only if $S \neq \varnothing$ and $\exists x \in \mathbb{Z}$ such that $\forall s \in S$, $x \leq s$.* By assumption, there exists $y \in G \setminus \{0\}$ since $0 \in G$ and $G \neq \{0\}$. There are two cases. If $y > 0$, then $y \in \mathbb{N}$, so we have found $y \in G \cap \mathbb{N} \setminus \{0\}$. Otherwise, $-y \in \mathbb{N}$, so we have found $-y \in G \cap \mathbb{N} \setminus \{0\}$. Therefore, $G \cap \mathbb{N} \setminus \{0\}$ has a well-defined minimum.

(2) We recall the Euclidean division lemma. *Given two integers a and b with $b \neq 0$, there exist unique integers q and r such that $a = bq + r$ and $0 \leq r < |b|$.* Now take arbitrary $x \in G$, then we have that $x = mq + r$ for some $(q, r) \in \mathbb{Z} \times [0, |m| - 1]$. Note that m is the smallest positive integer in $G \cap \mathbb{N} \setminus \{0\}$, so $r = 0$. This means that $x = mq$ for some $q \in \mathbb{Z}$. Up till now we have checked that $G \subseteq m\mathbb{Z}$. Now take arbitrary $x \in m\mathbb{Z}$, there exists $y \in \mathbb{Z}$ such that $x = my$. Now since $m \in G$ and G is a group, clearly $x \in G$ (add $m \in G$ for y times). This has shown that $m\mathbb{Z} \subseteq G$, so we can conclude that $G = m\mathbb{Z}$ and the proof is complete.

□

Example 1.48. Let A be a commutative group and $a_1, \cdots, a_r \in A$. Show that $\langle a_1, \cdots, a_r \rangle = \{a_1^{m_1} \cdots a_r^{m_r};\ m_i \in \mathbb{Z}\}$.

Proof. $\supseteq$ All a_i's belong to $\langle a_1, \cdots, a_r \rangle$, and since $\langle a_1, \cdots, a_r \rangle$ is stable under "multiplication" and inverse, clearly $a_1^{m_1} \cdots a_r^{m_r} \in \langle a_1, \cdots, a_r \rangle$, $m_i \in \mathbb{Z}$.

$\subseteq$ Denote $H := \{a_1^{m_1} \cdots a_r^{m_r};\ m_i \in \mathbb{Z}\}$. Each $a_i = a_1^0 \cdots a_i^1 \cdots a_r^0 \in H$, so $\{a_1, \cdots, a_r\} \subseteq H$. Now we check that H is a group. Clearly $e = a_1^0 \cdots a_r^0 \in H$. Now take arbitrary $x, y \in H$ and assuming that $x = a_1^{x_1} \cdots a_r^{x_r}$ and $y = a_1^{y_1} \cdots a_r^{y_r}$, we can see that $x \cdot y = a_1^{x_1+y_1} \cdots a_r^{x_r+y_r} \in H$ and $x^{-1} = a_1^{-x_1} \cdots a_r^{-x_r} \in H$. This proves that H is a group, and in particular a group containing $\{a_1, \cdots, a_r\}$. Note that $\langle a_1, \cdots, a_r \rangle$ is the smallest group containing $\{a_1, \cdots, a_r\}$, so $\langle a_1, \cdots, a_r \rangle \subseteq H$. □

Example 1.49. Let G and G' be two finite groups with the same cardinality n. Prove that if $n = 2, 3$, we have that $G' \simeq G$ (*i.e.*, G' and G are isomorphic). Find a counterexample for $n = 4$.

Proof. For this question we make use of the Cayley table (Appendix A). If $n = 2$, we write the Cayley table for $G = \{e, x\}$ as

$$T_{e,x} = \begin{array}{c|c|c} \cdot & e & x \\ \hline e & e & x \\ \hline x & x & \end{array} \tag{16}$$

We can see that the $(2,2)$th entry of $T_{e,x}$ can only be e because each row or column must have distinct elements. This means that the Cayley table for $G' = \{e', x'\}$ will be of exactly the same form, so by Theorem A.4, $G \simeq G'$ when $n = 2$. Now if $n = 3$, we write the Cayley table for $G = \{e, x, y\}$ as

$$T_{e,x,y} = \begin{array}{c|c|c|c} \cdot & e & x & y \\ \hline e & e & x & y \\ \hline x & x & & \\ \hline y & y & & \end{array} \tag{17}$$

We can see that the $(2,2)$th entry of $T_{e,x,y}$ can only be y, because it cannot be x and the $(2,3)$th entry that is in the same column as it cannot be y. Therefore the $(2,3)$th and the $(3,2)$th entry must be e, and thus the $(3,3)$th entry must be x. The table is again uniquely determined, which means that the Cayley table for $G' = \{e', x', y'\}$ will be of exactly the same form, so again by Theorem A.4, $G \simeq G'$ when $n = 3$. Now if $n = 4$, we can write the Cayley tables for $G = \{e, x, y, z\}$ and $G' = \{e', z', y', z'\}$ respectively as

$$T_{e,x,y,z} = \begin{array}{c|c|c|c|c} \cdot & e & x & y & z \\ \hline e & e & x & y & z \\ \hline x & x & e & z & y \\ \hline y & y & z & e & x \\ \hline z & z & y & x & e \end{array} \qquad T_{e',z',y',z'} = \begin{array}{c|c|c|c|c} \cdot & e & x & y & z \\ \hline e & e & x & y & z \\ \hline x & x & y & z & e \\ \hline y & y & z & e & x \\ \hline z & z & e & x & y \end{array} \tag{18}$$

These two Cayley tables are clearly not isomorphic. By the contrapositive of Theorem A.4, we can conclude that $G \not\simeq G'$, and thus the above is a counterexample for the case $n = 4$. □

9/11 Lecture

1.7 Normal Subgroups

Definition 1.50. Let $(G, \cdot)$ be a group and $H \leq G$. One says that H is a **normal subgroup** of G and writes $H \trianglelefteq G$, if $xhx^{-1} \in H$ for all $x \in G$ and $h \in H$. An equivalent formulation of the definition is that $\iota_x(H) \subseteq H$ for all $x \in G$.

Remark 1.51. Let $H \leq G$, then $H \trianglelefteq G$ if and only if $\iota_x(H) = H$ for all $x \in G$. Indeed, if $\iota_x(H) = H$ necessarily means that $\iota_x(H) \subseteq H$, so the $\Longleftarrow$ direction is trivial. For the other direction, suppose that $H \trianglelefteq G$, then $\iota_x(H) \subseteq H$ for all $x \in G$. But then $x^{-1} \in G$, so that $\iota_x^{-1}(H) = \iota_{x^{-1}}(H) \subseteq H$. Hence $H \subseteq \iota_x(H)$, so the proof is complete.

Proposition 1.52. If $f \in \mathrm{Hom}(G, G')$ and $H' \trianglelefteq G'$, then $f^{-1}(H') \trianglelefteq G$.

Proof. We want to show that $xhx^{-1} \in f^{-1}(H')$ for all $x \in G$ and $h \in f^{-1}(H')$. To see this, we observe that

$$f(xhx^{-1}) = \underbrace{f(x)}_{\in G'}\underbrace{f(h)}_{\in H'}\underbrace{f(x^{-1})}_{\in G'} \underbrace{\in H'}_{\text{since } H' \trianglelefteq G'} . \tag{19}$$

Therefore, $xhx^{-1} \in f^{-1}(H')$ for all $x \in G$ and $h \in f^{-1}(H')$, so the proof is complete. □

Corollary 1.53. If $f \in \mathrm{Hom}(G, G')$, then $\mathrm{Ker}(f) \trianglelefteq G$.

Proof. We check that $\{e'\} \trianglelefteq G'$. Indeed, $xe'x^{-1} = xx^{-1} = e' \in \{e'\}$ for all $x \in G'$. Now since $\mathrm{Ker}(f) = f^{-1}(\{e'\})$, we can conclude directly by Proposition 1.52. □

Example 1.54. Since $\det \in \mathrm{Hom}(GL_n(K), K^*)$, and $SL_n(K) = \mathrm{Ker}(\det)$, we have that $SL_n(K) \trianglelefteq GL_n(K)$.

Definition 1.55. For $(G, \cdot)$ a group, we denote by $Z(G)$ or Z_G the set $Z(G) = \{z \in G;\ xz = zx, \forall x \in G\}$, called the **center of** G. It is the subset of G the elements of which commute with all $x \in G$.

Proposition 1.56. $Z(G) \leq G$ and if $H \leq Z(G)$, then H is commutative and $H \trianglelefteq G$. In particular, $Z(G) \trianglelefteq G$.

Proof. Clearly $e \in Z(G)$ because $ex = xe = x$ for any $x \in G$. Now take arbitrary $z, z_1, z_2 \in Z(G)$, then for any $x \in G$, we have that

$$xz^{-1} = (z^{-1}z)(xz^{-1}) = z^{-1}(zx)z^{-1} = z^{-1}(xz)z^{-1} = (z^{-1}x)(zz^{-1}) = z^{-1}x, \tag{20}$$

$$x(z_1z_2) = \underbrace{(xz_1)}_{\in G} z_2 = z_2(xz_1) = \underbrace{(z_2x)}_{\in G} z_1 = z_1(z_2x) = (z_1z_2)x. \tag{21}$$

Therefore, we have shown that $Z(G) \leq G$. Now further assume that $H \leq Z(G)$. Now for any $h_1, h_2 \in H$, clearly $h_1, h_2 \in Z(G)$, and thus $h_1h_2 = h_2h_1$ which implies that H is commutative. Furthermore, take arbitrary $h \in H$ and $x \in G$, we can see that $xhx^{-1} = (xh)x^{-1} = (hx)x^{-1} = h \in H$, indicating that H is stable under G-conjugation, and hence $H \trianglelefteq G$. The proof is thus complete. Note that in particular, $Z(G) \leq Z(G)$ so that $Z(G) \trianglelefteq G$. □

Remark 1.57. A group G is commutative if and only if $G = Z(G)$. The $\Longleftarrow$ direction is trivial, since then for all $x \in G$, it commutes with all $z \in G$, meaning that $xz = xz$ for all $x, z \in G$. The other direction is also trivial. Assume that G is commutative, then for any $x \in G$, $xz = zx$ for any $z \in G$ so $x \in Z(G)$. Clearly $Z(G) \subseteq G$, and thus $Z(G) = G$ and the proof is complete.

Corollary 1.58. If A is a commutative group, then all its subgroups are normal.

Remark 1.59. We can show that $Z(G) = \mathrm{Ker}(\iota)$, where we recall that $\iota : G \to \mathrm{Inn}(G)$ such that $x \mapsto \iota_x$. Indeed, for any $z \in Z(G)$, we have that $zx = xz$ for all $x \in G$. Then $\iota(z)(x) = \iota_z(x) = zxz^{-1} = xzz^{-1} = x$ for all $x \in G$, meaning that $\iota(z) = \mathrm{Id}_G$, the neutral element in $\mathrm{Inn}(G)$. Hence $z \in \mathrm{Ker}(\iota)$, implying that $Z(G) \subseteq \mathrm{Ker}(\iota)$. On the other hand, for any $z \in \mathrm{Ker}(\iota)$, we have that $\iota(z) = \mathrm{Id}_G$. This means that for any $x \in G$, we have

$$x = \iota(z)(x) = \iota_z(x) = zxz^{-1} \implies xz = zx. \tag{22}$$

Hence $z \in Z(G)$, and thus $\mathrm{Ker}(\iota) \subseteq Z(G)$. Both directions of inclusion hold, so $Z(G) = \mathrm{Ker}(\iota)$.

Remark 1.60. We can show that $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$. Indeed, for any $f \in \mathrm{Inn}(G)$ and $g \in \mathrm{Aut}(G)$, we can write $f = \iota_z$ for some $z \in G$. Then for any $x \in G$, we have that

$$\begin{aligned}(g \circ \iota_z \circ g^{-1})(x) &= g(\iota_z(g^{-1}(x))) = g(g^{-1}(x)z(g^{-1}(x))^{-1}) \\ &= g(g^{-1}(x)z\underbrace{g^{-1}(x^{-1})}_{g \in \mathrm{Aut}(G)}) = \underbrace{g(g^{-1}(x))g(z)g(g^{-1}(x^{-1}))}_{g \in \mathrm{Aut}(G)} = xg(z)x^{-1} = \iota_{g(z)}(x),\end{aligned} \tag{23}$$

which implies that $g \circ \iota_z \circ g^{-1} = \iota_{g(z)} \in \mathrm{Inn}(G)$. Clearly $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$, so that $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.

1.8 Quotient Groups

Remark 1.61. Let X be a set and $\sim$ be an equivalence relation on X, then we denote by $Cl(x) = \{y \in X;\ y \sim x\}$ the equivalence class of x. We recall that for $x, x' \in X$, either $Cl(x) \cap Cl(x') = \varnothing$ or $Cl(x) = Cl(x')$. Hence the equivalence classes form a partition of X.

Definition 1.62. We denote by $\frac{X}{\sim}$ the set of classes for $\sim$, which is called the **quotient set** of X by $\sim$. For each $C \in \frac{X}{\sim}$, we pick on element $x_C \in C$ (hence $C = Cl(x_C)$), then the set $R = \left\{x_C;\ C \in \frac{X}{\sim}\right\}$ is called the **set of representatives** for $\sim$. In particular X is the disjoint union $X = \coprod_{x \in \mathbb{R}} Cl(x)$.

Remark 1.63. A subset $R \subseteq X$ is a set of representatives for $\sim$ if and only if the mapping $Cl : R \to \frac{X}{\sim}$ is a bijection.

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Remark 1.64. Let G be a group. For $A \subseteq G$ and $x, y \in G$, we set the notations $xAy = \{xay;\ a \in A\}$, $xA = xAe = \{xa;\ a \in A\}$, and $Ay = eAy = \{ay;\ a \in A\}$. From now on, $H \leq G$.

Proposition 1.65. The relations $y_H \sim x \iff y \in Hx$ and $y \sim_H x \iff y \in xH$ are equivalence relations on G.

Proof. We prove that ${}_H\sim$ is an equivalence relation on G, and the proof for $\sim_H$ would be analogous. First, we have that $x = ex \in Hx$ so $x_H \sim x$, which means that ${}_H\sim$ is reflexive. Second, if $x_H \sim y$, then there exists $h \in H$ such that $y = hx$. Then $x = h^{-1}y$ where $h^{-1} \in H$ because H is a subgroup. Therefore, $x \in Hy$ and thus $y_H \sim x$, which means that ${}_H\sim$ is symmetric. Finally, if $x_H \sim y$ and $y_H \sim z$, then there exists $h_1, h_2 \in H$ such that $y = h_1x$ and $z = h_2y$. Hence $z = h_2(h_1x) = (h_2h_1)x$ where $h_2h_1 \in H$ because H is a subgroup. Therefore, $z \in Hx$ and thus $x_H \sim z$, which means that ${}_H\sim$ is transitive. Now we can conclude that ${}_H\sim$ is an equivalence relation on G. □

Remark 1.66. Note that xH is the equivalence class $Cl_H(x) = \{y \in G;\ y_H \sim x\}$, and Hx is the equivalence class ${}_HCl(x) = \{y \in G;\ y_H \sim x\}$.

Definition 1.67. We call xH the **right H-coset** of x, and Hx the **left H-coset** of x. We denote by G/H the set of right H-cosets $G/H = \frac{G}{\sim_H}$ and $H\backslash G$ the set of left H-cosets $H\backslash G = \frac{G}{{}_H\sim}$.

Example 1.68. Define $\mathrm{inv} : G \to G$ by $\mathrm{inv}(x) = x^{-1}$. We can easily see that $\mathrm{inv} \circ \mathrm{inv} = \mathrm{Id}$, and we can check that $\mathrm{inv}(xH) = Hx^{-1}$ and $\mathrm{inv}(Hx) = x^{-1}H$. We want to show that inv induces a bijection between G/H and $H\backslash G$. Indeed, $\mathrm{inv}_H : xH \mapsto \mathrm{inv}(xH) = Hx^{-1}$ is a mapping from G/H to $H\backslash G$, and $\mathrm{inv}^H : Hx \mapsto \mathrm{inv}(Hx) = x^{-1}H$ is a mapping from $H\backslash G$ to G/H. Moreover, inv_H and inv^H are inverses of each other.

Lemma 1.69. For all $x \in G$, the mapping $l_x : G \to G$ such that $g \mapsto xg$ induces a bijection between H and xH, and the mapping $r_x : G \to G$ such that $g \mapsto gx$ induces a bijection between H and Hx.

Proof. We only prove for l_x, then r_x would be analogous. Indeed, the mapping $l_x : G \to G$ is bijective with inverse $l_{x^{-1}}$, and moreover $xH = l_x(H)$, so the proof is trivial. □

Corollary 1.70. If $|H| < \infty$, then $|xH| = |H|$ and $|Hx| = |H|$ for all $x \in H$.

Theorem 1.71 (Lagrange's theorem)**.** Let G be a group and $H \leq G$ be a subgroup of G. If G is finite, then $|H|$ divides $|G|$ and $|G| = |G/H| \times |H|$, and also $|G| = |H\backslash G| \times |H|$.

Proof. Let $R \subseteq G$ be a set of representatives of $G/H = \frac{G}{\sim_H}$, then $G = \coprod_{r \in R} rH$. Therefore, $|G| = \sum_{r \in R} |rH|$. But by Corollary 1.70, we have that $|rH| = |H|$, and as a result $|G| = |R| \times |H|$. This implies that $|H|$ divides $|G|$, and moreover we conclude the result by observing that $|R| = |G/H|$. The other result follows an analogous proof and will not be explicitly shown here. □

Remark 1.72. If G is a finite group with prime cardinality p, then $G = \langle x \rangle$ for any $x \neq e$ in G. Indeed by Lagrange's theorem (Theorem 1.71), we know that $|\langle x \rangle|$ divides $|G|$ because $\langle x \rangle \leq G$, but since $|G|$ is prime, either $|\langle x \rangle| = 1$ or $|\langle x \rangle| = |G| = p$. However $|\langle x \rangle| \geq 2$ because it at least contains the neutral element e and $x \neq e$. Therefore, we can conclude that $G = \langle x \rangle$ because $\langle x \rangle \subseteq G$ but both have the same cardinality.

9/15 Recitation

Example 1.73. For G a group and $x, y \in G$, we denote by $[x, y]$ the element $[x, y] := xyx^{-1}y^{-1}$ of G. It is called the **commutator** of x and y. We denote by $D(G)$ or $[G, G]$ the **commutator subgroup**, also called the **derived subgroup** of G, which is by definition generated by all the commutators in G, *i.e.*,

$$D(G) = [G, G] := \langle [x, y];\ x, y \in G \rangle . \tag{24}$$

(1) Let $f \in \mathrm{Hom}(G, G')$, and suppose that $S \subseteq G$. Show that $f(\langle S \rangle) = \langle f(S) \rangle$.

(2) Suppose that S is stable under conjugation, *i.e.*, $\iota_x(S) \subseteq S$ for any $x \in G$, prove that $\langle S \rangle \trianglelefteq G$.

(3) Prove that $[G, G] \trianglelefteq G$.

(4) Prove that $[G, G] = \{e\}$ if and only if G is commutative.

Proof. (1) $\subseteq$ Take any $y \in f(\langle S \rangle)$, then there exists $x \in \langle S \rangle$ such that $y = f(x)$. By Proposition 1.20, either $x = e$ or $x = \prod_{i=1}^{n} s_i^{\epsilon_i}$ for some $n \in \mathbb{N}$, $\epsilon_i = \pm 1$, and $s_i \in S$. In the former case, $y = f(e) = e' \in \langle f(S) \rangle$. In the latter case, since f is a group homomorphism, we have that $y = f\left(\prod_{i=1}^{n} s_i^{\epsilon_i}\right) = \prod_{i=1}^{n} f(s_i)^{\epsilon_i}$. This means that $y \in \langle f(S) \rangle$ because $f(s_i) \in f(S)$ and $\epsilon_i = \pm 1$ and we can apply the other direction of Proposition 1.20.

$\supseteq$ Take any $y \in \langle f(S) \rangle$, then by Proposition 1.20, either $y = e'$ or $y = \prod_{i=1}^{n} s_i'^{\epsilon_i}$ for some $n \in \mathbb{N}$, $\epsilon_i = \pm 1$, and $s_i' \in f(S)$. In the former case, we note that $f(e) = e' = y$ so $y \in f(\langle S \rangle)$. In the latter case, there exists $s_1, \cdots, s_n \in S$, such that $f(s_i) = s_i'$ for any $1 \leq i \leq n$. Since f is a group homomorphism, we have that $y = \prod_{i=1}^{n} f(s_i)^{\epsilon_i} = f\left(\prod_{i=1}^{n} s_i^{\epsilon_i}\right)$. This means that $y \in f(\langle S \rangle)$ because $s_i \in S$ and $\epsilon_i = \pm 1$ and we can apply the other direction of Proposition 1.20 to see that $\prod_{i=1}^{n} s_i^{\epsilon_i} \in \langle S \rangle$.

(2) For any $x \in G$, we know that $\iota_x \in \mathrm{Inn}(G) \subseteq \mathrm{Hom}(G, G)$, so $\iota_x(\langle S \rangle) = \langle \iota_x(S) \rangle$ for any $S \subseteq G$ as is shown in the previous part. But since $\iota_x(S) \subseteq S$, we have that $\langle \iota_x(S) \rangle \subseteq \langle S \rangle$, so $\iota_x(\langle S \rangle) \subseteq \langle S \rangle$. This, by the alternative definition, concludes that $\langle S \rangle \trianglelefteq G$.

(3) For any $s \in [G, G]$, there exists $x, y \in G$ such that $s = xyx^{-1}y^{-1}$. Then for any $g \in G$, we have that

$$\begin{aligned} gsg^{-1} = gxyx^{-1}y^{-1}g^{-1} = gx(g^{-1}g)y(g^{-1}g)x^{-1}(g^{-1}g)y^{-1}g^{-1} &= (gxg^{-1})(gyg^{-1})(gx^{-1}g^{-1})(gy^{-1}g^{-1}) \\ &= \underbrace{(gxg^{-1})}_{\in G}\underbrace{(gyg^{-1})}_{\in G}(gxg^{-1})^{-1}(gyg^{-1})^{-1}, \end{aligned} \tag{25}$$

so $\iota_g(s) \in [G, G]$ for any $s \in [G, G]$ and $g \in G$. In other words, $\iota_g([G, G]) \subseteq [G, G]$ for any $g \in G$. By the previous part, we can thus conclude that $\langle [G, G] \rangle \trianglelefteq G$.

(4) $\Longrightarrow$ If $[G, G] = \{e\}$, then for any $x, y \in G$, we have that $xyx^{-1}y^{-1} = e$. In other words, $xy = yx$ so that G is commutative.

$\Longleftarrow$ If G is commutative, then for any $x, y \in G$, we have that $xyx^{-1}y^{-1} = (xx^{-1})(yy^{-1}) = e$, so $[G, G] = \{e\}$.

The proof is thus complete. □

Example 1.74. Let K be a field. For $1 \leq i \neq j \leq n$ and $x \in K$, set $E_{i,j}(x) := I_n + xE_{i,j} \in \mathfrak{M}_n(K)$, where $E_{i,j}$ has all entries equal to 0 except the entry in position (i, j) which is equal to 1. For $t \in K^*$, set $d_t = \mathrm{diag}(t, I_{n-1}) \in \mathfrak{M}_n(K)$.

(1) Check that $E_{i,j}(x) \in SL_n(K)$ and $d_t \in GL_n(K)$.

(2) Prove that $SL_2(K) = \langle \{E_{i,j}(x);\ 1 \leq i \neq j \leq 2,\ x \in K\} \rangle$.

(3) Prove that $SL_n(K) = \langle \{E_{i,j}(x);\ 1 \leq i \neq j \leq n,\ x \in K\} \rangle$.

(4) Prove that $GL_n(K) = \langle \{E_{i,j}(x);\ 1 \leq i \neq j \leq n,\ x \in K\} \cup \{d_t;\ t \in K^*\} \rangle$.

Proof. (1) Note that $E_{i,j}(x)$ can either be upper triangular or lower triangular with diagonal elements all 1 because for $i \neq j$. Therefore, $\det(E_{i,j}(x)) = 1$ which means that $E_{i,j}(x) \in SL_n(K)$. On the other hand, d_t is diagonal with diagonal elements $t, 1, \cdots, 1$, so that $\det(d_t) = t \neq 0$ for $t \neq 0$. This means that $d_t \in GL_n(K)$.

(2) Ask the Professor...

(3) Ask the Professor...

(4) Ask the Professor...

□

9/18 Lecture

Lemma 1.75. Let $H \leq G$, then the following are equivalent.

(1) $H \trianglelefteq G$.

(2) $gHg^{-1} = H$ for all $g \in G$.

(3) $gH = Hg$ for all $g \in G$.

Proof. We first argue that (1) and (2) are equivalent. (2) implies (1) by definition. Now assume that $H \trianglelefteq G$, so that $gHg^{-1} \subseteq H$ for all $g \in G$. Taking $g^{-1} \in G$, we also have that $g^{-1}Hg \subseteq H$ for all $g \in G$, which necessarily means that $H \subseteq gHg^{-1}$. With both inclusions, we can conclude that $gHg^{-1} = H$ for all $g \in G$ and thus (1) implies (2) as well. Now we move on to show that (2) is equivalent to (3). Indeed, multiplying by g on the right of (2) gives (3), and multiplying by g^{-1} on the right of (3) gives (2). In conclusion, the three statements are all equivalent, and the proof is complete. □

Remark 1.76. In particular, if $H \trianglelefteq G$, then $G/H = H\backslash G$. By convention, we denote $\frac{G}{H} := G/H = H\backslash G$, and we set $\overline{g}$ for $gH = Hg \in \frac{G}{H}$.

Lemma 1.77. Suppose that $H \trianglelefteq G$, then the multiplication mapping $m : \frac{G}{H} \times \frac{G}{H} \to \frac{G}{H}$, such that $m(\overline{x}, \overline{y}) = \overline{xy}$ is well-defined.

Proof. It suffices to check that if $\overline{x'} = \overline{x}$ and $\overline{y'} = \overline{y}$, then $\overline{x'y'} = \overline{xy}$. By assumption, there exists $x' = xh$ for some $h \in H$ and $y' = yk$ for some $k \in H$. As a result, we can see that $x'y' = xhyk = x(yy^{-1})hyk = xy((y^{-1}hy)k)$. Note that since $H \trianglelefteq G$, we have that $y^{-1}hy \in H$ because $y \in \frac{G}{H} \subseteq G$. Hence $\overline{x'y'} = \overline{xy}$ and the proof is complete. □

Remark 1.78. From the previous lemma, we can see that it makes sense to put $\overline{g} \cdot \overline{g'} = \overline{gg'}$.

Theorem 1.79. If $H \trianglelefteq G$, then $\left(\frac{G}{H}, \cdot\right)$ is a group with neutral element $\overline{e} = H$.

Proof. This is trivial by previous observations and will be omitted here. □

Proposition 1.80. If $H \trianglelefteq G$, then the **canonical projection/surjection** $p : G \to \frac{G}{H}$ defined by $p(g) = \overline{g}$, is a surjective homomorphism with kernel $\mathrm{Ker}(p) = H$.

Proof. Since $p(xy) = \overline{xy} = \overline{x} \cdot \overline{y} = p(x)p(y)$, clearly p is a homomorphism. The surjectivity of p is trivial. Now if $x \in \mathrm{Ker}(p)$, then $p(x) = \overline{e} = H$. Necessarily $x \in H$, since otherwise $\overline{x}$ has at least an element not in H. On the other hand, if $x \in H$, then $p(x) = \overline{x} = H = \overline{e}$, so that $x \in \mathrm{Ker}(p)$. By both directions of inclusion, we can conclude that $\mathrm{Ker}(p) = H$, and the proof is complete. □

Corollary 1.81. If $H = \{e\}$, then p is an isomorphism between G and $\frac{G}{\{e\}}$.

Proof. By the previous proposition, p is a surjective homomorphism with $\mathrm{Ker}(p) = H$, so that $\mathrm{Ker}(p) = \{e\}$, which necessarily means that p is injective as well. Therefore, p is bijective and thus an isomorphism. The proof is now complete. □

9/20 Lecture

Example 1.82. We recall that if G is commutative, then all its subgroups are normal. Also, we recall from recitations that the subgroups of $\mathbb{Z}$ are the groups $n\mathbb{Z}$, $n \in \mathbb{N}$. We can see that

$$\frac{\mathbb{Z}}{n\mathbb{Z}} \begin{cases} \simeq \mathbb{Z}, & \text{if } n = 0, \\ = \left\{\overline{0}, \cdots, \overline{n-1}\right\}, & \text{if } n \geq 1. \end{cases} \tag{26}$$

In the case $n \geq 1$, we have that $\left|\frac{\mathbb{Z}}{n\mathbb{Z}}\right| = 1$, and $R = \{0, \cdots, n-1\}$ is a system of representatives for $\frac{\mathbb{Z}}{n\mathbb{Z}}$.

Theorem 1.83 (First isomorphism theorem)**.** For $f \in \mathrm{Hom}(G, G')$ and $p : G \to \frac{G}{\mathrm{Ker}(f)}$ the canonical projection, there exists a unique $\overline{f} : \frac{G}{\mathrm{Ker}(f)} \to G'$, such that $f = \overline{f} \circ p$, *i.e.*, such that the following diagram commutes.

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "p"'] & G' \\ \frac{G}{\mathrm{Ker}(f)} \arrow[ur, "\overline{f}"'] & \end{tikzcd}$$

Moreover, $\overline{f} \in \mathrm{Hom}\left(\frac{G}{\mathrm{Ker}(f)}, G'\right)$ and is injective.

Proof. If $f = \overline{f} \circ p$ and $\overline{x} = p(x) \in \frac{G}{\mathrm{Ker}(f)}$, then the relation $\overline{f}(\overline{x}) = \overline{f}(p(x)) = (\overline{f} \circ p)(x) = f(x)$ shows that $\overline{f}$ is unique if it exists. Therefore, it suffices to prove that such $\overline{f}$ exists, *i.e.*, to check that the mapping $\overline{f} : \frac{G}{\mathrm{Ker}(f)} \to G'$ such that $\overline{x} \mapsto f(x)$ is well-defined. Indeed, if $\overline{x'} = \overline{x}$, then there exists $k \in \mathrm{Ker}(f)$, such that $x' = xk$. As a consequence, $f(x') = f(xk) = f(x)f(k) = f(x)$ because f is a homomorphism and k is in the kernel. This proves that $\overline{f}$ is well-defined. Now we prove that $\overline{f}$ is a group homomorphism. For any $\overline{x}, \overline{x'} \in \frac{G}{\mathrm{Ker}(f)}$, we have that

$$\overline{f}(\overline{x} \cdot \overline{x'}) = \overline{f}(\overline{xx'}) = f(xx') = \underbrace{f(x)f(x')}_{f \in \mathrm{Hom}(G,G')} = \overline{f}(\overline{x})\overline{f}(\overline{x'}), \tag{27}$$

so indeed $\overline{f} \in \mathrm{Hom}\left(\frac{G}{\mathrm{Ker}(f)}, G'\right)$. Finally if $\overline{x} \in \mathrm{Ker}(\overline{f})$, we have that $\overline{f}(\overline{x}) = e' \in G'$. This means that $f(x) = e'$, so $x \in \mathrm{Ker}(f)$. Note again that the quotient space is $\frac{G}{\mathrm{Ker}(f)}$, and thus $\overline{x} = \overline{e}$. Therefore, $\mathrm{Ker}(\overline{f}) \subseteq \{\overline{e}\}$. But $\overline{e} \in \mathrm{Ker}(\overline{f})$, so necessarily $\mathrm{Ker}(\overline{f}) = \{\overline{e}\}$, which implies that $\overline{f}$ is injective, so the proof is complete. □

Corollary 1.84. The mapping $\overline{f}$ induces an isomorphism between $\frac{G}{\mathrm{Ker}(f)}$ and $\mathrm{Im}(f) = f(G)$. In other words,

$$\frac{G}{\mathrm{Ker}(f)} \overset{\overline{f}}{\simeq} \mathrm{Im}(f). \tag{28}$$

In particular for finite G, we have that $|G| = |\mathrm{Ker}(f)| \times |\mathrm{Im}(f)|$.

Proof. This is a direct consequence of the first isomorphism theorem (Theorem 1.83), *i.e.*, the quotient space of G divided by $\mathrm{Ker}(f)$ is isomorphic to $\mathrm{Im}(f)$ if f is a group homomorphism from G. Moreover if G is finite, we have that $|G|/|\mathrm{Ker}(f)| = \left|\frac{G}{\mathrm{Ker}(f)}\right| = |\mathrm{Im}(f)|$ by the Lagrange's theorem (Theorem 1.71), and thus $|G| = |\mathrm{Ker}(f)| \times |\mathrm{Im}(f)|$. The proof is now complete. □

Theorem 1.85. Let G be a **cyclic group**, *i.e.*, a group generated by one element, up to isomorphism. Then there exists a unique $n \in \mathbb{N}$, such that $G \simeq \frac{\mathbb{Z}}{n\mathbb{Z}}$. If $n = 0$ then $G \simeq \mathbb{Z}$ is infinite, and if otherwise $n \geq 1$, G is finite of cardinality n.

Proof. Let $G = \langle x \rangle$, then the mapping $f : \mathbb{Z} \to G$, defined by $f(k) = x^k$, is a surjective homomorphism. Indeed, we have that $f(k_1 + k_2) = x^{k_1+k_2} = x^{k_1}x^{k_2} = f(k_1)f(k_2)$ for any $k_1, k_2 \in \mathbb{Z}$ and f is clearly surjective. Since $\mathrm{Ker}(f) \leq \mathbb{Z}$, it is of the form $n\mathbb{Z}$. Therefore by Corollary 1.84 and surjectivity of f, we can conclude that $\frac{\mathbb{Z}}{n\mathbb{Z}} = \frac{\mathbb{Z}}{\mathrm{Ker}(f)} = \mathrm{Im}(f) = G$. The rest of the theorem holds by Example 1.82, so the proof is complete. □

Example 1.86. (1) For $n \geq 1$, we check that $\frac{\mathbb{Z}}{n\mathbb{Z}} \simeq \mathbb{U}_n := \{z \in \mathbb{C};\ z^n = 1\}$. Consider the mapping $\phi : \mathbb{Z} \to \mathbb{U}_n$, such that $k \mapsto (\exp(i2\pi/n))^k$, which is a surjective homomorphism. Indeed, we have that $\phi(k_1 + k_2) = (\exp(i2\pi/n))^{k_1+k_2} = (\exp(i2\pi/n))^{k_1}(\exp(i2\pi/n))^{k_2} = \phi(k_1)\phi(k_2)$ for any $k_1, k_2 \in \mathbb{Z}$ and ϕ is clearly surjective because it maps to all powers of the nth root of unity. Since $\mathrm{Ker}(\phi) \leq \mathbb{Z}$, it is of the form $n\mathbb{Z}$. Therefore by Corollary 1.84 and surjectivity of ϕ, we can conclude that $\frac{\mathbb{Z}}{n\mathbb{Z}} = \frac{\mathbb{Z}}{\mathrm{Ker}(\phi)} \simeq \mathrm{Im}(\phi) = \mathbb{U}_n$ for some $n \in \mathbb{N}$.

(2) We check that $\frac{\mathbb{R}}{\mathbb{Z}} \simeq \mathbb{C}_u := \{z \in \mathbb{C};\ |z| = 1\}$. Consider the mapping $\phi : \mathbb{R} \to \mathbb{C}_u$, such that $x \mapsto \exp(i2\pi x)$, which is a surjective homomorphism. Indeed, we have that $\phi(x_1+x_2) = \exp(i2\pi(x_1+x_2)) = \exp(i2\pi x_1)\exp(i2\pi x_2) = \phi(x_1)\phi(x_2)$ for any $x_1, x_2 \in \mathbb{R}$ and ϕ is clearly surjective because it maps to the whole unit circle. Now we check that $\mathrm{Ker}(\phi) = \mathbb{Z}$. Indeed, if $x \in \mathrm{Ker}(\phi)$, we necessarily have that $\phi(x) = \exp(i2\pi x) = 1$, the rightmost

point of the unit circle in the complex plane. Hence $2\pi x \in 2\pi\mathbb{Z}$, *i.e.*, $x \in \mathbb{Z}$, which implies that $\mathrm{Ker}(\phi) \subseteq \mathbb{Z}$. The other direction of inclusion is trivial, so we can conclude that $\mathrm{Ker}(\phi) = \mathbb{Z}$. Now by Corollary 1.84 and surjectivity of ϕ, we can conclude that $\frac{\mathbb{R}}{\mathbb{Z}} = \frac{\mathbb{Z}}{\mathrm{Ker}(\phi)} \simeq \mathrm{Im}(\phi) = \mathbb{C}_u$.

(3) We check that $\frac{GL_n(K)}{SL_n(K)} \simeq K^*$. Recall that $GL_n(K)$ is the group of all $n \times n$ invertible matrices and $SL_n(K)$ is the group of all $n \times n$ matrices with determinant 1 (and thus invertible). Consider the mapping $\det : GL_n(K) \to K^*$, which is a surjective homomorphism. Indeed, we have that $\det(M_1 M_2) = \det(M_1)\det(M_2)$ for any $M_1, M_2 \in GL_n(K)$ and det is clearly surjective because K^* does not include 0 determinant and we can always find invertible matrices with any non-zero determinant. Now recall that $\mathrm{Ker}(\det) = SL_n(K)$, which is trivial. Therefore by Corollary 1.84 and surjectivity of det, we can conclude that $\frac{GL_n(K)}{SL_n(K)} = \frac{GL_n(K)}{\mathrm{Ker}(\det)} \simeq \mathrm{Im}(\det) = K^*$.

9/22 Recitation

To be done...

9/24 Recitation

To be done...

9/25 Lecture

Lemma 1.87. Take $f \in \mathrm{Hom}(G, G')$. If $K \trianglelefteq G'$ then $f^{-1}(K) \trianglelefteq G$, whereas if $H \trianglelefteq G$ then $f(H) \trianglelefteq f(G)$.

Proof. For the first part, take arbitrary $x \in f^{-1}(K)$ and $x' \in G$. Then we have that $x \in G$ and thus $x'x(x')^{-1} \in G$. By arbitrariness of x and x', we can conclude that $f^{-1}(K) \trianglelefteq G$. Now for the second part, take arbitrary $y \in f(H)$ and $y' \in f(G)$, there exists $h \in H$ and $g \in G$ such that $y = f(h)$ and $y' = f(g)$. Then we have that $y'y(y')^{-1} = f(g)f(h)(f(g))^{-1} = f(ghg^{-1}) \in f(H)$ because f is a group homomorphism and $H \trianglelefteq G$. Hence by arbitrariness of g and h, we can conclude that $f(H) \trianglelefteq f(G)$. The proof is now complete. $\square$

Remark 1.88. Though $H \trianglelefteq G$ implies $f(H) \trianglelefteq f(G)$, it does *not* imply that $f(H) \trianglelefteq G'$ in general.

Theorem 1.89. If $f \in \mathrm{Hom}(G, G')$, then the mapping $H \mapsto f(H)$ is a bijection from the set $\{H;\ \mathrm{Ker}(f) \leq H \leq G\}$ to the set $\{K;\ K \leq \mathrm{Im}(f)\}$ of subgroups of $\mathrm{Im}(f)$, with inverse $K \mapsto f^{-1}(K)$. Moreover, this bijection and its inverse preserve normal subgroups.

Proof. Denote $\mathscr{A} = \{H;\ \mathrm{Ker}(f) \leq H \leq G\}$ and $\mathscr{B} = \{K;\ K \leq \mathrm{Im}(f)\}$. Also denote the mappings $m : H \mapsto f(H)$ and $n : K \mapsto f^{-1}(K)$. We want to show that m and n are bijective and are inverses of each other. First we show that $m \circ n = \mathrm{Id}_{\mathscr{B}}$, *i.e.*, for any $K \in \mathscr{B}$, we claim that $K = m(n(K)) = f(f^{-1}(K))$.

- $\subseteq$ Fix arbitrary $k \in K$. Since $K \leq \mathrm{Im}(f)$, we have that $k \in \mathrm{Im}(f)$, so by definition of image there exists $x \in G$ such that $f(x) = k$. Then by definition of preimage $x \in f^{-1}(K)$, and thus $k = f(x) \in f(f^{-1}(K))$.
- $\supseteq$ Fix arbitrary $k \in f(f^{-1}(K))$, then there exists $h \in f^{-1}(K)$ such that $k = f(h)$. By definition of preimage, we can see that $k = f(h) \in K$.

Next we show that $n \circ m = \mathrm{Id}_{\mathscr{A}}$, *i.e.*, for any $H \in \mathscr{A}$, we claim that $H = n(m(H)) = f^{-1}(f(H))$.

- $\subseteq$ Fix arbitrary $h \in H$, then $f(h) \in f(H)$. By definition of preimage, $h \in f^{-1}(f(H))$.
- $\supseteq$ Fix arbitrary $h \in f^{-1}(f(H))$, then $f(h) \in f(H)$ by definition of preimage. This means that there exists $h' \in H$, such that $f(h) = f(h')$. Since f is a group homomorphism, we have that $f(h(h')^{-1}) = f(h)(f(h'))^{-1} = e'$, so $h(h')^{-1} \in \mathrm{Ker}(f)$. Since $\mathrm{Ker}(f) \leq H$, we have that $h(h')^{-1} \in H$, and thus $h \in Hh' = \{xh';\ x \in H\}$. But we claim that $Hh' = H$.
  - $\subseteq$ Fix arbitrary $h \in Hh'$, then there exists $x \in H$ such that $h = xh'$. But $x \in H$ and $h' \in H$, so $h = xh' \in H$.

$\supseteq$ Fix arbitrary $h \in H$, then take $x = h(h')^{-1}$. Indeed $x \in H$ since $h \in H$ and $h' \in H$, and clearly $h = xh'$ so that $h \in Hh'$.

Therefore, we can see that $h \in Hh' = H$.

Up till now, we have shown that $m \circ n = \mathrm{Id}_{\mathscr{B}}$ and $n \circ m = \mathrm{Id}_{\mathscr{A}}$, which prove that m and n are inverses of each other, and thus automatically bijective. Moreover, they both preserve normal subgroups by Lemma 1.87, so the proof is complete. $\square$

Corollary 1.90 (Subgroups of quotients)**.** Assume that $N \trianglelefteq G$ and $p \in \mathrm{Hom}\left(G, \frac{G}{N}\right)$ the canonical projection, then the mapping $H \mapsto p(H) = \frac{H}{N}$ is a bijection from the set $\{H;\ N \leq H \leq G\}$ to the set $\left\{K;\ K \leq \frac{G}{N}\right\}$, with inverse $K \mapsto p^{-1}(K)$. Moreover, this bijection and its inverse preserve normal subgroups.

Proof. This is a direct consequence of the previous theorem by taking $G = G$ and $G' = \frac{G}{N}$. Note that $\mathrm{Ker}(p) = N$. $\square$

Remark 1.91. In the corollary above, if $N \leq H$, then automatically $N \trianglelefteq H$ because $N \trianglelefteq G$. Hence $\frac{H}{N}$ is indeed a group.

Example 1.92. Take $n \geq 1$, then we can check that the subgroups of $\frac{\mathbb{Z}}{n\mathbb{Z}}$ are exactly those of the form $\frac{d\mathbb{Z}}{n\mathbb{Z}}$ where $d \in D(n) = \{d \in \mathbb{N}^*;\ d \mid n\}$. First of all, if $n\mathbb{Z} \leq H \leq \mathbb{Z}$, then H must be of the form $d\mathbb{Z}$ because $H \leq \mathbb{Z}$ and necessarily $d \mid n$ because it needs to contain all elements of $n\mathbb{Z}$. Taking $N = n\mathbb{Z}$ and $G = \mathbb{Z}$ in Corollary 1.90, we can see that the mapping $m : \{d\mathbb{Z};\ d \in D(n)\} \to \left\{K;\ K \leq \frac{\mathbb{Z}}{n\mathbb{Z}}\right\}$, such that $d\mathbb{Z} \mapsto p(d\mathbb{Z}) = \frac{d\mathbb{Z}}{n\mathbb{Z}}$ is a bijection. Clearly all $\frac{d\mathbb{Z}}{n\mathbb{Z}}$ with $d \in D(n)$ would then be subgroups of $\frac{\mathbb{Z}}{n\mathbb{Z}}$. Moreover by the surjectivity of m, we can then see that for any $K \leq \frac{\mathbb{Z}}{n\mathbb{Z}}$, there exists $d\mathbb{Z}$ with $d \mid n$, such that $K = m(d\mathbb{Z}) = \frac{d\mathbb{Z}}{n\mathbb{Z}}$. We can thus conclude that the subgroups of $\frac{\mathbb{Z}}{n\mathbb{Z}}$ are exactly those of the form $\frac{d\mathbb{Z}}{n\mathbb{Z}}$ with $d \in D(n)$.

9/27 Lecture

Proposition 1.93 (Digression on cyclic groups)**.** A subgroup of a cyclic group is cyclic.

Proof. Let G be a cyclic group and assume that it is generated by a. Also let H be an arbitrary subgroup of G. If $H = \{e\}$, then $H = \langle e \rangle$ and thus is cyclic so we are done. Therefore we assume that $H \neq \{e\}$. This means that $a^n \in H$ for some $n \in \mathbb{N}^*$. Let m be the smallest integer in $\mathbb{N}^*$ such that $a^m \in H$, then we claim that $H = \langle a^m \rangle$. To see this, we need to show that every $b \in H$ is some positive integer power of a^m. Now since $b \in H$ and $H \leq G$, there exists $n \in \mathbb{N}$ such that $b = a^n$. Then by Euclidean division property, there exists unique integers q and r such that $n = mq + r$, with $0 \leq r < m$. Consequently, $b = a^{mq+r} = (a^m)^q \cdot a^r$, and thus $a^r = b \cdot (a^m)^{-q}$. We note that $b \in H$ and $a^m \in H$, so that $a^r \in H$. But m is the smallest integer in $\mathbb{N}^*$ such that $a^m \in H$, so necessarily $r = 0$ (otherwise $a^r \in H$ with $0 < r < m$ leads to a contradiction). Therefore, each b can be written as $b = (a^m)^q$, which means that $H = \{a^m\}$. By arbitrariness of H, we can hence conclude that every subgroup of a cyclic group G is cyclic, and the proof is complete. $\square$

Alternative proof. If G is cyclic, it is isomorphic to $\frac{\mathbb{Z}}{n\mathbb{Z}}$ for some $n \in \mathbb{N}$ (Theorem 1.85). Hence, it is sufficient to prove that the statement holds for subgroups H of $\frac{\mathbb{Z}}{n\mathbb{Z}}$. If $n = 0$, then $\frac{\mathbb{Z}}{n\mathbb{Z}} \simeq \mathbb{Z}$, so any subgroup H must be of the form $H = m\mathbb{Z} = \langle m \rangle$ for some $m \in \mathbb{N}$, thus being cyclic. Otherwise if $n \geq 1$, we have just shown in a previous example that any subgroup H must be of the form $H = \frac{d\mathbb{Z}}{n\mathbb{Z}} = \langle \overline{d} \rangle$ for some $d \in D(n)$, also being cyclic. The proof is thus complete. $\square$

Example 1.94. We shall check that for $d \in D(n)$, we have that $\frac{d\mathbb{Z}}{n\mathbb{Z}} \simeq \frac{\mathbb{Z}}{\frac{n}{d}\mathbb{Z}}$. How to do this?

Theorem 1.95 (Second isomorphism theorem)**.** Take $f \in \mathrm{Hom}(G, G')$, and fix $N \trianglelefteq G$ with $N \subseteq \mathrm{Ker}(f)$. If p is the canonical projection from G to $\frac{G}{N}$, then there is a unique mapping $\overline{f} : \frac{G}{N} \to G'$ such that $f = \overline{f} \circ p$. Moreover, $\overline{f}$ is a group homomorphism such that $\mathrm{Im}(\overline{f}) = \mathrm{Im}(f)$ and $\mathrm{Ker}(\overline{f}) = \frac{\mathrm{Ker}(f)}{N}$.

Proof. If $\overline{f}$ exists, it is unique by the formula $\overline{f}(\overline{g}) = \overline{f}(p(g)) = (\overline{f} \circ p)(g) = f(g)$. Therefore, it suffices to prove that such $\overline{f}$ exists, *i.e.*, to check that the mapping $\overline{f} : \frac{G}{N} \to G'$ such that $\overline{g} \mapsto f(g)$ is well-defined. Indeed, if $\overline{g'} = \overline{g}$, then there exists $n \in N$ such that $g' = gn$. As a consequence, $f(g') = f(gn) = f(g)f(n) = f(g)$ because f is a homomorphism and $n \in N \subseteq \mathrm{Ker}(f)$. This proves that $\overline{f}$ is well-defined. Now we prove that $\overline{f}$ is a group homomorphism. For any $\overline{g}, \overline{g'} \in \frac{G}{N}$, we have that

$$\overline{f}(\overline{g} \cdot \overline{g'}) = \overline{f}(\overline{gg'}) = f(gg') = \underbrace{f(g)f(g')}_{f \in \mathrm{Hom}(G,G')} = \overline{f}(\overline{g})\overline{f}(\overline{g'}), \tag{29}$$

so indeed $\overline{f} \in \mathrm{Hom}\left(\frac{G}{N}, G'\right)$. Finally, $\mathrm{Im}(\overline{f}) = \mathrm{Im}(f)$ directly follows from the formula $\overline{f}(\overline{g}) = f(g)$, and we note that

$$\overline{g} \in \mathrm{Ker}(\overline{f}) \iff \overline{f}(\overline{g}) = e' \iff f(g) = e' \iff g \in \mathrm{Ker}(f) \iff \overline{g} \in \frac{\mathrm{Ker}(f)}{N}. \tag{30}$$

This necessarily means that $\mathrm{Ker}(\overline{f}) = \frac{\mathrm{Ker}(f)}{N}$, and the proof is now complete. □

Corollary 1.96. With the same notations as in Theorem 1.95, we have that

$$\frac{\frac{G}{N}}{\frac{\mathrm{Ker}(f)}{N}} \overset{\overline{(\overline{f})}}{\simeq} \mathrm{Im}(f). \tag{31}$$

Proof. By the second isomorphism theorem (Theorem 1.95), we have that $\overline{f} \in \mathrm{Hom}\left(\frac{G}{N}, G'\right)$. By Corollary 1.84, we then have that

$$\frac{\frac{G}{N}}{\mathrm{Ker}(\overline{f})} \overset{\overline{(\overline{f})}}{\simeq} \mathrm{Im}(\overline{f}). \tag{32}$$

Again by the second isomorphism theorem (Theorem 1.95), we have that $\mathrm{Ker}(\overline{f}) = \frac{\mathrm{Ker}(f)}{N}$ and $\mathrm{Im}(\overline{f}) = \mathrm{Im}(f)$, so the proof is now complete by substituting these results into the relation above. □

Corollary 1.97. Suppose that N and H are two normal subgroups of G with $N \subseteq H$, then $\frac{H}{N}$ is a normal subgroup of $\frac{G}{N}$ and we have that

$$\frac{\frac{G}{N}}{\frac{H}{N}} \simeq \frac{G}{H}. \tag{33}$$

Proof. Since H is a normal subgroup of G, clearly $\frac{H}{N}$ is a normal subgroup of $\frac{G}{N}$. Let $f := p_H$ be the canonical projection from G to $\frac{G}{H}$. Then by the previous corollary we can conclude that

$$\frac{\frac{G}{N}}{\frac{H}{N}} = \frac{\frac{G}{N}}{\frac{\mathrm{Ker}(f)}{N}} \simeq \mathrm{Im}(f) = \frac{G}{H}. \tag{34}$$

The proof is thus complete. □

Example 1.98. If $d \in D(n)$, then $d\mathbb{Z}$ and $n\mathbb{Z}$ are both normal subgroups of $\mathbb{Z}$ and $n\mathbb{Z} \subseteq d\mathbb{Z}$. Then $\frac{d\mathbb{Z}}{n\mathbb{Z}}$ is a normal subgroup of $\frac{\mathbb{Z}}{n\mathbb{Z}}$, and that

$$\frac{\frac{\mathbb{Z}}{n\mathbb{Z}}}{\frac{d\mathbb{Z}}{n\mathbb{Z}}} \simeq \frac{\mathbb{Z}}{d\mathbb{Z}}. \tag{35}$$

A Cayley Table

Let G be a finite group with r elements. To a numbering $g_1, \cdots, g_r$ of the elements of G, one can associate the corresponding **multiplication table** of G, also known as the **Cayley table**.

$$T_{g_1,\cdots,g_r} := \begin{array}{c|c|c|c|c|c} \cdot & g_1 & g_2 & \cdots & g_{r-1} & g_r \\ \hline g_1 & g_1^2 & g_1 \cdot g_2 & \cdots & g_1 \cdot g_{r-1} & g_1 \cdot g_r \\ \hline g_2 & g_2 \cdot g_1 & g_2^2 & \cdots & g_2 \cdot g_{r-1} & g_2 \cdot g_r \\ \hline \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \hline g_{r-1} & g_{r-1} \cdot g_1 & g_{r-1} \cdot g_2 & \cdots & g_{r-1}^2 & g_{r-1} \cdot g_r \\ \hline g_r & g_r \cdot g_1 & g_r \cdot g_2 & \cdots & g_r \cdot g_{r-1} & g_r^2 \end{array} \tag{36}$$

In other words, the table is of size $r \times r$, and the entry in position (i, j) contains the value $g_i \cdot g_j$. Often one chooses $g_1 = e$, the neutral element, in the numbering of G.

Example A.1. We write the multiplication table of $\mathbb{U}_5 = \{1, u, u^2, u^3, u^4\}$ where $u = \exp(2i\pi/5)$, the 5th root of unity. Using the relation $u^5 = 1$, we obtain

$$T_{1,u,u^2,u^3,u^4} = \begin{array}{c|c|c|c|c|c} \cdot & 1 & u & u^2 & u^3 & u^4 \\ \hline 1 & 1 & u & u^2 & u^3 & u^4 \\ \hline u & u & u^2 & u^3 & u^4 & 1 \\ \hline u^2 & u^2 & u^3 & u^4 & 1 & u \\ \hline u^3 & u^3 & u^4 & 1 & u & u^2 \\ \hline u^4 & u^4 & 1 & u & u^2 & u^3 \end{array} \tag{37}$$

Proposition A.2. Let G be a finite group with r elements and $g_1, \cdots, g_r$ be a numbering of the elements of G. Then each row and each column of $T_{g_1,\cdots,g_r}$ contains distinct elements, which are exactly all elements of G in a possibly different order.

Proof. We observe that for a fixed $g \in G$, the mapping $l_g : G \to G$ such that $x \mapsto g \cdot x$ is bijective with inverse mapping $l_{g^{-1}}$. The same holds for $r_g : x \mapsto x \cdot g$ such that $x \mapsto x \cdot g$, which is bijective with inverse mapping $r_{g^{-1}}$. This implies that the elements of the ith row are $l_{g_i}(g_1), \cdots, l_{g_i}(g_r)$ (from left to right), and the elements of the jth column are $r_{g_i}(g_1), \cdots, r_{g_i}(g_r)$ (from top to bottom). The proof is thus complete. □

Definition A.3. Let G and G' be two finite groups of the same cardinality r, with respective elements numbering $g_1, \cdots, g_r$ and $g'_1, \cdots, g'_r$. We say that $T := T_{g_1,\cdots,g_r}$ and $T' = T_{g'_1,\cdots,g'_r}$ are **isomorphic** if for all $i, j \in \{1, \cdots, r\}$, the following holds: if the entry $T_{i,j} = g_i \cdot g_j$ is equal to g_k (for some $k \in \{1, \cdots, r\}$), then the entry $T'_{i,j} = g'_i \cdot g'_j$ is equal to g'_k (for the same k). We denote this as $T' \simeq T$.

Theorem A.4. Let G and G' be two finite groups, then they are isomorphic if and only if they have the same cardinatlity, say r, and if there are numberings $g_1, \cdots, g_r$ of G and $g'_1, \cdots, g'_r$ of G', such that $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$.

Proof. $\Longrightarrow$ If G and G' are isomorphic, then there exists $\phi \in \mathrm{Iso}(G, G')$. Fix an arbitrary numbering $g_1, \cdots, g_r$ of G, and set $g'_i = \phi(g_i)$. Then $g'_1, \cdots, g'_r$ is a numbering of G', and we observe that if $g_i \cdot g_j = g_k$, then $g'_i \cdot g'_j = \phi(g_i) \cdot \phi(g_j) = \phi(g_i \cdot g_j) = \phi(g_k) = g'_k$. Hence $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$.

$\Longleftarrow$ Assume that there are numberings $g_1, \cdots, g_r$ of G and $g'_1, \cdots, g'_r$ of G', such that $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$. We define the mapping $\phi : G \to G'$ such that $\phi(g_i) = g'_i$ for $i = 1, \cdots, r$. Clearly ϕ is a bijection and for any i, j, take the unique integer $1 \leq k \leq r$ such that $g_k = g_i \cdot g_j$, we can deduce that

$$\phi(g_i \cdot g_j) = \phi(g_k) = g'_k = g'_i \cdot g'_j = \phi(g_i) \cdot \phi(g_j). \tag{38}$$

Therefore, ϕ preserves multiplication and is thus a group homomorphism, and further since ϕ is a bijection, we can conclude that ϕ is an isomorphism. G and G' are thus isomorphic.

□